



Modelling Financial Markets

Chapter 5: Conditional Volatility — GARCH Models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-MIDAS: GARCH with Mixed Data Sampling (long-run component)
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GW: Giacomini–White (test)

Acronyms used in this material (2/2)

- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ PIT: Probability Integral Transform
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile–Quantile (plot)
- ▣ QSD: Quasi Score-Driven (model)
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

Chapter Route and Outcomes

Three questions

- How does the variance evolve? ARCH, GARCH, persistence, forecasts (Part I)
- How are the models estimated and checked? Likelihood, QMLE inference, boundary tests, asymmetry, diagnostics (Part I)
- Do the forecasts improve? Out-of-sample losses, tests for many models, limits and extensions, a case study (Part II)
- Technical detail sits in the [Appendix](#) at the end

After this chapter you can

- estimate GARCH, GJR and EGARCH with heavy-tailed innovations and read every parameter
- attach valid uncertainty to persistence, half-life and tests on the boundary of the parameter space
- diagnose standardised residuals and compare variance forecasts with proxy-robust losses

Volatility Comes in Bursts

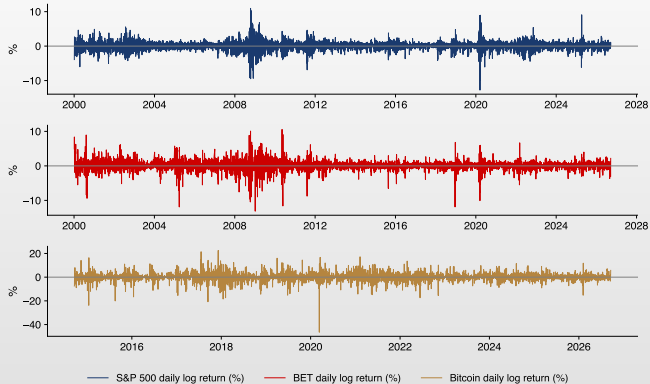


Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

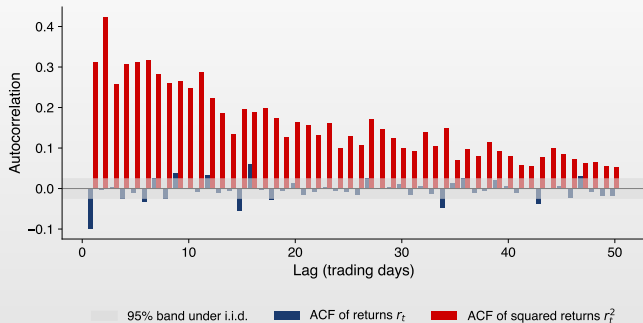
- Crashes are not confined to isolated days: large moves are followed by large moves
 - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of [Chapter 1](#)
- Volatility enters every risk number: Value at Risk ([Chapter 7](#)), portfolio weights ([Chapter 4](#)), option prices
- Question: can we turn the clustering into a **forecast** of tomorrow's risk, and conduct valid inference on it?

Volatility Clusters in Every Market



- Daily log returns $r_t = 100 (\ln P_t - \ln P_{t-1})$ in %, closes up to 18 September 2026; S&P 500 and BET from 2000, Bitcoin from 17 September 2014
- Equity indices: weekdays; days whose close equals the previous close (holidays filled with the last price) are dropped: 3 for the S&P 500, 0 for the BET; Bitcoin: 7 days a week
- Quiet years alternate with turbulent months in all three: 2008, March 2020, April 2025

The Sign Is Unpredictable, the Size Is Not



- S&P 500, lag 1: -0.099 for r_t , 0.313 for r_t^2 ; all 50 autocorrelations of r_t^2 up to lag 50 lie outside ± 0.024
- Heavy tails too: excess kurtosis 10.7 (S&P 500), 11.1 (BET), 12.0 (Bitcoin); 0 for the Normal distribution
- Object to model: $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t - 1$

Testing for ARCH Effects

- H_0 : no clustering, $\text{Var}(r_t | \mathcal{F}_{t-1})$ constant
- **McLeod–Li** [McLeod & Li \(1983\)](#): Ljung–Box $Q(10)$ on r_t^2 ; reject if $Q(10) > 18.3$ (χ_{10}^2 , 5%)
- **Engle's LM test** [Engle \(1982\)](#): regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$, $\hat{\varepsilon}_t = r_t - \bar{r}$
 - ▶ $LM = nR^2$; H_0 : all five slopes are 0; reject if $LM > 11.1$ (χ_5^2 , 5%)
- Every statistic is far above its critical value (all $p < 0.001$): clustering in equities, crypto, a managed exchange rate and gold; weakest for Bitcoin and gold
- One-sided score tests and the full table: [▶ Appendix](#)

Series	$Q(10)$ on r^2	LM(5)	$\hat{\rho}_1(r^2)$
S&P 500	6133	1684	0.313
BET	2440	982	0.330
BET-TR	714	371	0.171
Bitcoin	213	115	0.137
EUR/RON	2420	871	0.342
Gold	744	329	0.110

The ARCH Model of Engle (1982)

- ▣ Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
 - ▶ Conditional variance $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$; unconditional variance $\sigma^2 = \mathbb{E}\sigma_t^2$
- ▣ **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2$ Engle (1982)
 - ▶ $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- ▣ A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ▣ ARCH(1): unconditional variance $\sigma^2 = \omega/(1 - \alpha_1)$, finite if $\alpha_1 < 1$; its fourth moment is the case $\beta = 0$ of the GARCH result below
- ▣ Engle received the 2003 Nobel Prize for ARCH (history: [Appendix](#))

From ARCH to GARCH (Bollerslev, 1986)

- ▣ Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- ▣ **GARCH(1,1)** Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0, \alpha \geq 0, \beta \geq 0$; α : reaction to news; β : memory of past variance
- ▣ Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
 - ▶ An ARCH(∞) with geometrically declining weights, using only three parameters
- ▣ GARCH(p, q) adds more lags; in practice GARCH(1,1) is the workhorse Hansen & Lunde (2005)

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$
 - ▶ An ARMA(1,1) in ε_t^2 with autoregressive coefficient $\alpha + \beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
 - ▶ $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
 - ▶ S&P 500, GARCH(1,1) with Normal innovations: $\rho_1 = 0.356$, $\rho_{10} = 0.298$; sample 0.313 and 0.248
- $\alpha + \beta$ is therefore called the **persistence** of volatility

Stationarity and the Unconditional Variance

- ▣ **Covariance stationarity:** $\alpha + \beta < 1$; then

- ▶ $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$

- ▣ Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$

- ▣ **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite

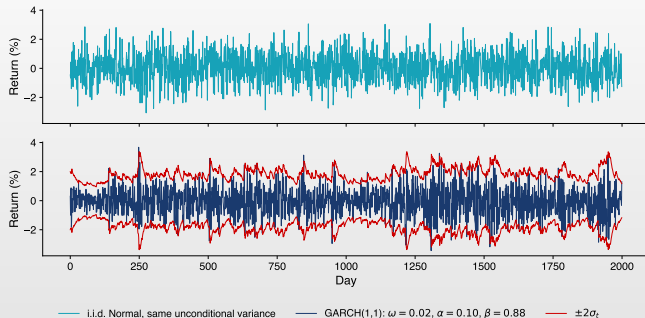
- ▶ Yet, if $\alpha > 0$ and z_t^2 is non-degenerate, the process is still **strictly** stationary: the condition $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ **Nelson (1990)** holds by Jensen, since $\mathbb{E} \ln(\alpha z_t^2 + \beta) < \ln(\alpha + \beta) = 0$

- ▣ Variance targeting: fix $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$ with the sample variance $\hat{\sigma}^2$

Fourth Moment and Kurtosis

- With Normal z_t , the kurtosis of ε_t is $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$
 - ▶ Finite only if $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (Seminar 1, Exercise A9)
 - ▶ ARCH(1) is the case $\beta = 0$: $K = 3(1 - \alpha^2)/(1 - 3\alpha^2)$, finite only if $\alpha < 1/\sqrt{3}$: heavy tails even with Normal innovations
- S&P 500, GARCH(1,1) with Normal innovations: $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$
 - ▶ $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.990435$: just below 1, so $K = 12.1$; sample kurtosis of the returns: 13.7
- K is ill-conditioned near the moment boundary: report it from unrounded estimates (a rounding example: [Appendix](#); Seminar 5, A4)
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, the same unconditional variance (1), and the bands $\pm 2\sigma_t$
- Quiet stretches and bursts appear only in the GARCH path
- Normal innovations, yet fat-tailed returns: a mixture of Normal distributions with varying variances has heavy tails (kurtosis on a long path: [Appendix](#))

Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$
 - ▶ The gap to the long-run variance shrinks by the factor $\alpha + \beta$ each day
- **Half-life:** number of days after which half of a variance shock has gone
 - ▶ $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples: $\alpha + \beta = 0.95$: 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in $\alpha + \beta$ near 1 imply very different half-lives: estimation uncertainty matters (next section; Seminar 5, A3 and B7)

EWMA and IGARCH: No Mean Reversion

- ▣ **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data and zero-mean returns
 - ▶ Weights $(1 - \lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
 - ▶ EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: no parameter to estimate
- ▣ Multi-step forecasts at the close of day t differ
 - ▶ EWMA: flat, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2$ for every h
 - ▶ IGARCH with $\omega > 0$: rising, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2 + (h - 1)\omega$
 - ▶ Stationary GARCH: pulled towards σ^2 at the rate $\alpha + \beta$
- ▣ EWMA serves as a benchmark in the forecast comparison

Maximum Likelihood

- Parameters $\theta = (\mu, \omega, \alpha, \beta)$; given θ , the recursion produces $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$
 - ▶ Each day contributes the density of ε_t given the past: the likelihood factorises in time
- No closed form: numerical optimisation under $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- Start-up: σ_1^2 is set to the sample variance (its effect fades at rate β)
- Scale: returns in %, so that the variance is of order 1
 - ▶ EUR/RON (daily s.d. about 0.28%) is fitted on 10 r_t : unscaled, the optimiser stopped at a local optimum

Gaussian QMLE: the Result and Its Assumptions

- **Gaussian QMLE:** maximise the Normal likelihood even if z_t is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- **Consistency** (GARCH(1,1)) [Berkas, Horváth & Kokoszka \(2003\)](#), [Francq & Zakoïan \(2004\)](#)
 - ▶ Strict stationarity $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$; θ_0 in a compact set; z_t^2 non-degenerate; no moment condition on ε_t
- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E} z_t^4 < \infty$
 - ▶ $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \rightarrow N(0, (\kappa_z - 1)J^{-1})$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$
- Sandwich $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$; the inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
 - ▶ A : minus the expected Hessian; B : variance of the score; robust / classical s.e. = $\sqrt{(\kappa_z - 1)/2}$, 1 under Normality
- Variants (estimated mean, IGARCH, explosive case): [Appendix](#)

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
 - ▶ Theory: robust / classical s.e. = $\sqrt{(4.73 - 1)/2} = 1.37$; observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t (suffix -t: Student-t innovations) gives $\hat{\nu} = 3.18 < 4$: under that law $\mathbb{E}z_t^4 = \infty$, and the condition for Gaussian-QMLE normality fails
 - ▶ Residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample; s.e. ratios 2.51 and 2.01
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$, to a non-Gaussian limit: its robust t -statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014), or the Student-t MLE (bounded score: no fourth moment needed if the law is right), with bootstrap inference near bounds

Heavy-Tailed Innovations

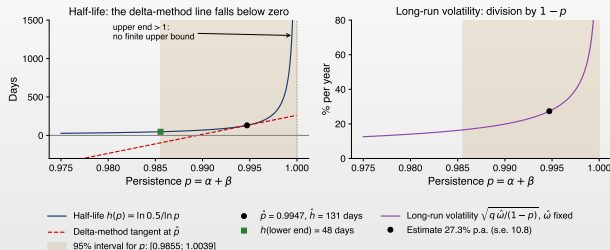
- **Student-t** [Bollerslev \(1987\)](#): $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$, unit variance for $\nu > 2$
 - ▶ ν (degrees of freedom): the smaller ν , the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter $\lambda \in (-1, 1)$
 - ▶ $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
 - ▶ $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust s.e.	Classical s.e.	Ratio
ω	0.0157			
α	0.123	0.011	0.010	1.06
β	0.872	0.010	0.010	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Daily returns in %, 2000–2026; mean $\hat{\mu} = 0.079\%$ per day; full table with every standard error: Seminar 5, B1
- $\hat{\beta}$: most of today's variance is yesterday's variance; $\hat{\alpha}$: news adds a moderate update; $\hat{\nu} = 6.26$: tails clearly heavier than those of the Normal distribution
- Half-life 131 days; long-run volatility $\sqrt{251.4 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$ p.a. ($q = 251.4$ returns a year), above the sample 19.2%: dividing by $1 - \hat{\alpha} - \hat{\beta}$ makes it imprecise (next slide)

Near-Unit Persistence: What the Delta Method Can and Cannot Do



- Delta method: $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$; long-run volatility 27.3% p.a., s.e. 10.8 points
- Half-life: $\partial h / \partial p = 24\,789$ at $\hat{p} = 0.994712$, $\widehat{\text{s.e.}}(\hat{p}) = 0.0047$, so $\widehat{\text{s.e.}}(\hat{h}) \approx 116$ days, and $\hat{h} \pm 1.96 \widehat{\text{s.e.}}$ includes negative half-lives: the linearisation fails
- Transform the interval for p instead: $[0.9855; 1.0039]$ gives $h \in [48; \infty)$; whether $p = 1$ is plausible needs a boundary test (next slides)

Testing on the Boundary: a Decision Table

Situation under H_0	Example	Law of $LR = 2(\ell_1 - \ell_0)$	5% critical value
Interior restriction	GJR (asymmetric GARCH) vs GARCH, $\gamma = 0$ (with $\alpha > 0$)	χ_1^2	3.84
One parameter on the boundary, the others identified and interior	Normal vs Student-t ($1/\nu = 0$); IGARCH ($\alpha + \beta = 1$)	$\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ Self & Liang (1987), Andrews (2001)	2.71
A nuisance parameter unidentified	No ARCH: $\alpha = 0$ leaves β unidentified	non-standard Andrews (2001)	simulate
Several bounds active at once	GJR with $\hat{\alpha} = 0$, test of $\gamma = 0$	non-standard	simulate

- Non-nested models: **AIC** = $-2\ell + 2k$, **BIC** = $-2\ell + k \ln n$ (k parameters); lower is better
- A Wald t for an estimate at the bound (e.g. $\hat{\alpha} = 0$) is not $N(0, 1)$; a parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary Andrews (2000): simulate under H_0 instead

Boundary Tests on Our Data

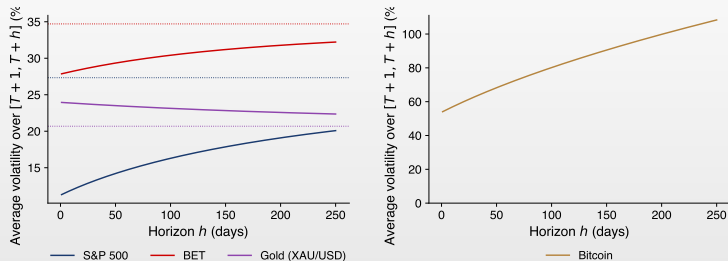
- Normal vs Student-t innovations, GARCH(1,1): $LR = 311.0$ (S&P 500), 1082.0 (Bitcoin)
 - ▶ Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- GJR-t (GJR: an asymmetric GARCH, next section), S&P 500: $\hat{\alpha} = 0.000$; its t -test is not standard, and the χ^2_1 law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$
- IGARCH, $H_0: \alpha + \beta = 1$ (GARCH-t): $LR = 1.11$ (S&P 500), 1.14 (BET)
 - ▶ S&P 500: $p = 0.29$ (χ^2_1), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 , $B = 999$)
 - ▶ BET: $p = 0.28$, 0.14 and 0.17
- Under H_0 , 52% of the re-estimates have $\hat{\alpha} + \hat{\beta} \geq 0.999$ (at the bound $\alpha + \beta = 1$), as the mixture predicts: a unit root in variance cannot be rejected for either index

Worked Example: One-Day and Ten-Day Forecasts

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}_T^2 = 0.565$
 - ▶ $\hat{\sigma}_{T+1}^2 = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_T^2 + \hat{\beta}\hat{\sigma}_T^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 2.97$; day 10: $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$
 - ▶ $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$, volatility 2.38%
 - ▶ The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}_{T+1}^2 < \hat{\sigma}^2$ and volatility is forecast to rise

 MFM_ch5_garch_estimation

Volatility Term Structure



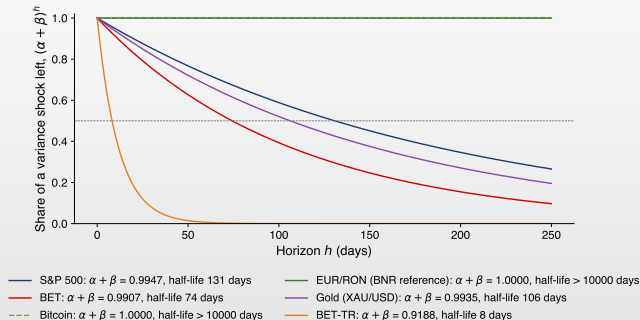
- Average volatility over the next h days, forecast from 18.09.2026 (GARCH(1,1)-t); dotted: long-run level
- S&P 500 rises from 11.3% to 20.1% in a year (below its long run today); gold falls from 23.9% to 22.4%
- Bitcoin, EUR/RON: IGARCH with $\hat{\omega} > 0$, the variance forecast grows by $\hat{\omega}$ per day (Bitcoin: 108.3% in a year): an artefact of the unit-root estimate

GARCH(1,1)-t in Six Markets

Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

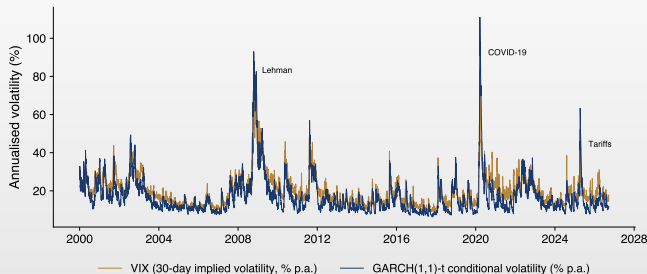
- Long-run volatility in % p.a., each series annualised with its own frequency; – when $\hat{\alpha} + \hat{\beta} \geq 0.999$; EUR/RON $\hat{\omega}$ is 0.0000 at four decimals
- Equities and gold: persistence 0.99–0.995; the BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
- Different samples: BET-TR starts in 2014, so its lower persistence is not comparable with the BET 2000–2026; Bitcoin: $\hat{\nu} < 4$, an infinite fourth moment under the fitted law
- EUR/RON, a managed rate: on 6 May 2025 a move of +1.20% after days of $\pm 0.004\%$ gives a standardised residual of 154

Persistence: How Fast Do Shocks Fade?



- $(\hat{\alpha} + \hat{\beta})^h$: the share of a variance shock left after h days; dashed line: one half
- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8; in Bitcoin and EUR/RON shocks never fade in the model
- Level shifts in volatility can mimic high persistence [Lamoureux & Lastrapes \(1990\)](#) (regime switching: [Hamilton & Susmel \(1994\)](#)): report half-lives as intervals (Seminar 5, B7)

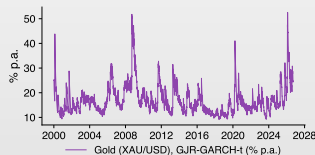
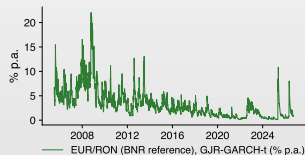
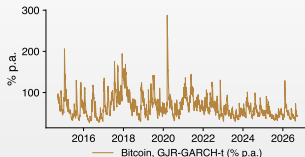
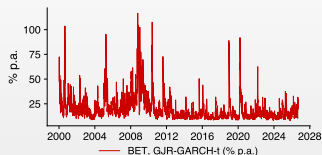
S&P 500: GARCH Volatility and the VIX



- $\hat{\sigma}_t \sqrt{q}$ ($q = 251.4$) from GARCH(1,1)-t and the VIX (30-day implied volatility), 2000–2026: correlation 0.89
- The VIX is higher on average: 19.8% against 17.0%, above GARCH on 84% of the days
- Not a measured variance risk premium: the premium compares VIX^2 with a 30-day variance forecast under the real-world probability; $\hat{\sigma}_t$ is a one-day volatility

 MFM_ch5_sp500_vix

BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t volatility, each series annualised with its own frequency (Bitcoin: 365 days)
- BET: 2008 dominates, now 26.4%; Bitcoin: 50–100% is normal, the highest peak is March 2020
- EUR/RON: a managed calm with bursts in 2022 and 2025, poorly described by a daily GARCH; gold: a new high of 53% in 2026

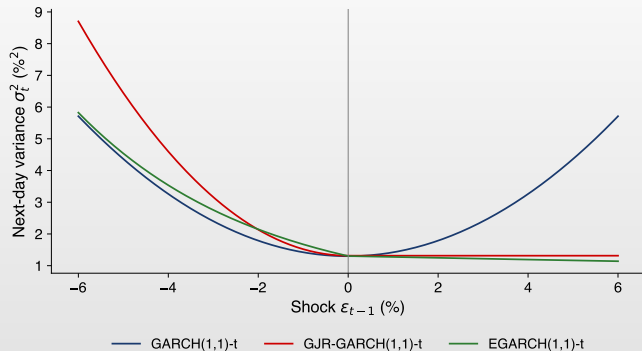
GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- Bad news raises future volatility more than good news of the same size (stylised fact 9, [Chapter 1](#)); GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ [Glosten, Jagannathan & Runkle \(1993\)](#)
 - ▶ Good news: slope α ; bad news: slope $\alpha + \gamma$; positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$ (γ may be negative)
 - ▶ Persistence $\alpha + \gamma/2 + \beta$ for symmetric z_t ; for skewed z_t replace $1/2$ by $\mathbb{E}[z_t^2 1(z_t < 0)]$ (Seminar 5, A2)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$; symmetry test: $H_0: \gamma = 0$ by t or by LR against GARCH(1,1)
- Candidate mechanisms, not tested here: leverage (a falling price raises the debt-to-equity ratio) and volatility feedback (higher expected volatility raises required returns, so prices fall now)

EGARCH (Nelson, 1991)

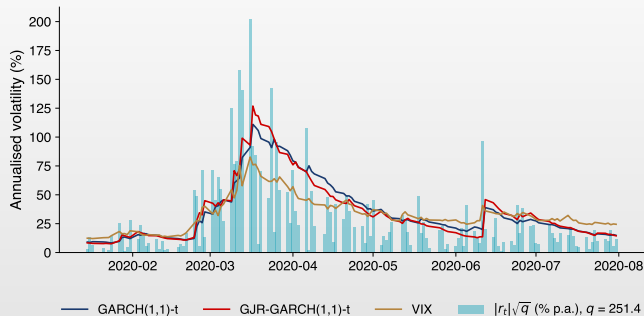
- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ Nelson (1991)
 - ▶ Size effect α , sign effect γ ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ ($t = -15.2$), $\hat{\beta} = 0.979$
- In arch, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve for the S&P 500



- News impact curve Engle & Ng (1993): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed at the sample variance 1.47 (%²)
- Shock of -3% : GJR 3.16, EGARCH 2.76; $+3\%$: GJR 1.31, EGARCH 1.22; GARCH: 2.40 for both
- After bad news the forecast is 2.4 times (GJR) and 2.3 times (EGARCH) that after good news: ratios of levels, not of increases

Zoom: The COVID-19 Crash, 2020



- January–July 2020: $|r_t|$ annualised, GARCH-t and GJR-GARCH-t volatility, VIX
- 24 February ($r_t = -3.41\%$): the VIX went from 17.1% to 25.0% the same day; GARCH-t from 12.2% to 22.5% only the next day: GARCH reacts, markets price at once
- After $r_t = -12.8\%$ on 16 March, peaks on 17 March: GARCH-t 111%, GJR-t 127%, higher because the shocks were negative

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- Student-t innovations; robust t; $LR \sim \chi^2_1$ (3.84) if the Student-t density is correct; the robust t is $N(0, 1)$ only if the other parameters are interior (not for $\hat{\alpha} = 0$)
- Bad news weighs more (GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$): S&P 500 strongly ($LR = 235.7$), BET and BET-TR weakly (4.9, 10.3), EUR/RON slightly; Bitcoin: none detected ($LR = 0.7$); gold: inverted ($\hat{\gamma}_{GJR} = -0.043$)
- Leverage for equities and a safe-haven role for gold are interpretations consistent with these signs, not tested mechanisms

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0, 1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

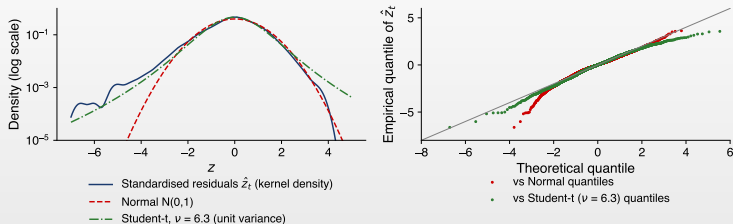
- ΔBIC : $\text{BIC} = -2\ell + k \ln n$ minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: Student-t instead of Normal innovations lowers BIC by about 300; asymmetry lowers it by a further 230–270; winner: EGARCH with skewed-t innovations
- BIC ranks in-sample fit, not forecast accuracy: the forecast section tests whether these gains survive out of sample

 MFM_ch5_garch_estimation

Standardised Residuals and the Sign-Bias Tests

- **Standardised residual:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- Checks, each with H_0 : the model captures that feature
 - ▶ Mean equation: Ljung–Box $Q(10)$ on $\hat{z}_t$; variance equation: $Q(10)$ on $\hat{z}_t^2$ and LM(5); distribution: QQ plot against the assumed density
- Sign-bias regression [Engle & Ng \(1993\)](#): $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18 in Engle and Ng, 1993)
 - ▶ b_1 : sign bias; b_2 , b_3 : negative and positive size bias; joint test $nR^2 \sim \chi_3^2$ (7.81 at 5%), conservative because it omits the gradient of the fitted variance

S&P 500: Are the Innovations Normal?



- Standardised residuals of GARCH(1,1) with Normal innovations: excess kurtosis falls from 10.7 (returns) to 1.73; skewness -0.53, Jarque-Bera 1153
- 15 residuals below -4: 0.21 expected under the Normal, 8.5 under a Student-t with $\nu = 6.3$; largest: $\hat{z} = -6.6$ (27 February 2007)
- A longer left tail: skewed-t with $\hat{\lambda} = -0.150$ (GJR, $\hat{\nu} = 7.7$ degrees of freedom), hence BIC prefers skewed-t innovations

S&P 500: Diagnostics of Five Models

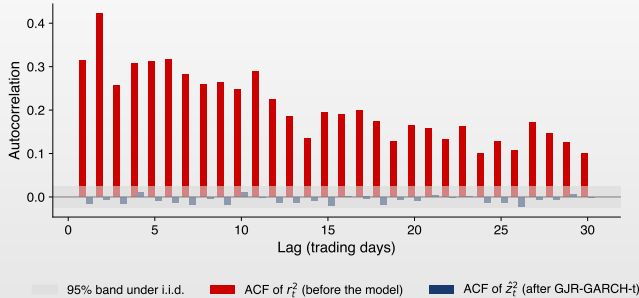
Model	Q(10) on $\hat{z}$	Q(10) on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	3.64	41.8
GARCH-t	21.9	13.6	5.8	3.56	40.7
GJR-t	22.3	10.2	4.5	3.61	28.1
EGARCH-t	27.6	10.5	8.2	2.04	9.9
GJR-skewed-t	19.7	10.7	5.0	3.91	31.8

- 5% critical values: 18.3 (χ^2_{10}), 11.1 (χ^2_5), 7.81 (χ^2_3), $|t| > 1.96$; naive laws, conservative with estimated parameters (next slide)
- Mean: Q(10) on $\hat{z}$ of 20–28 in every model: a little autocorrelation that a constant mean ignores
- Variance: Q(10) on $\hat{z}^2$ falls from 6133 (on r_t^2) to 10–16 and LM(5) is not significant: these naive tests detect no clustering left
- Asymmetry: symmetric GARCH fails the joint test (40.7), GJR still fails (28.1), EGARCH comes closest (9.9, $p = 0.020$); with $n = 6\,714$, judge economic size, not only p-values

Residual Diagnostics Have Estimated Parameters

- ▣ $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ_m^2 Li & Mak (1994); the fitted variance has already absorbed part of the dependence at small lags
- ▣ S&P 500, Gaussian QMLE of GARCH(1,1), $Q(10)$ on $\hat{z}_t^2$
 - ▶ Naive Ljung–Box: 16.4, $p = 0.089$: not rejected at 5%
 - ▶ Corrected for the estimated parameters: 19.1, $p = 0.038$: rejected at 5%, the verdict flips
- ▣ The naive test is conservative: it rejects a misspecified variance equation too rarely; the joint sign-bias test carries the same caveat
- ▣ The correction matrix and its entries: [▶ Appendix](#)

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500, lags 1–30: autocorrelations of r_t^2 (before the model) and of $\hat{z}_t^2$ from GJR-GARCH-t
- After the model they sit inside the i.i.d. band: no remaining variance dynamics are detected at these lags
- The band is the naive one: with estimated parameters it is too wide at small lags (previous slide)

Diagnostics in Other Markets: a Decision Table

Market	Residual pattern	Possible failure	Next check
BET	$Q(10)$ on $\hat{z} = 116.6$	autocorrelation in the mean (thin trading, Chapter 2)	AR(1) mean equation
Bitcoin	$Q(10)$ on $\hat{z}^2 = 4.7$, joint sign bias 1.3: no rejection	$\hat{\nu} = 3.18 < 4$: $\text{Var}(z_t^2) = \infty$, so the χ^2 calibration is not justified	tests on $ \hat{z}_t $
EUR/RON	excess kurtosis 3190	one residual of 154 (May 2025) dominates	inspect the largest residuals; a bounded-score model
Gold	EGARCH: $Q(10)$ on $\hat{z}^2 = 48.8$	clustering left despite a good BIC	choose by diagnostics, not by BIC alone

- Always look at the largest residuals: one observation can dominate every diagnostic

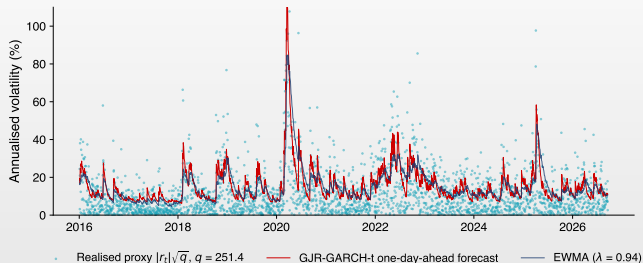
The Proxy Problem and Robust Losses

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g. r_t^2
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
 - ▶ Its R^2 against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
 - ▶ S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
 - ▶ Robust: MSE $(r_t^2 - h_t)^2$ and **QLIKE** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast
 - ▶ Not robust: losses on volatility $|r_t| - \sqrt{h_t}$ or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

Mincer–Zarnowitz and Diebold–Mariano

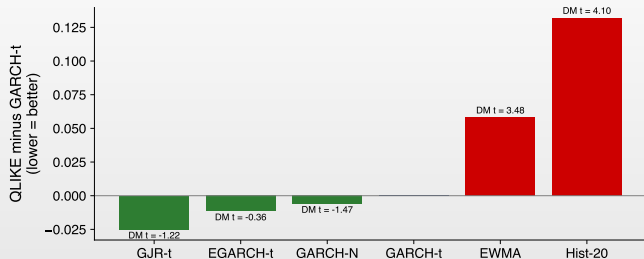
- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$
 - ▶ Unbiased forecast: $a = 0$, $b = 1$ (joint Wald test with HAC errors); R^2 measures information content
 - ▶ $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
 - ▶ $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$ with a HAC standard error; $\approx N(0, 1)$ under equal accuracy; $DM < -1.96$: A significantly better
 - ▶ The $N(0, 1)$ law assumes given forecasts; for estimated, nested models (GARCH-t inside GJR-t) it can fail, so DM t is read descriptively there (GW slide below)

Out-of-Sample Design and Forecast Paths, 2016–2026



- S&P 500: 2 692 one-day forecasts; refit each January on all data since 2000 (expanding window), daily updates with fixed parameters
- Timing: h_t uses data up to the close of day $t - 1$ and is compared with r_t^2 ; changing r_t must leave h_t unchanged
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t; benchmarks: EWMA and Hist-20 (20-day historical variance); shown: GJR-t, EWMA, $|r_t|\sqrt{q}$ ($q = 251.4$)

S&P 500: Does Anything Beat GARCH(1,1)?



- Mean QLIKE minus that of GARCH-t, with the DM t (HAC, 5 lags); lowest QLIKE: GJR-t (0.732)
- GARCH family: DM t from -1.47 to -0.36, descriptive for these nested pairs
- EWMA and Hist-20 are clearly worse ($t = 3.48, 4.10$): mean reversion and estimated parameters pay off [Hansen & Lunde \(2005\)](#)
- MZ: unbiasedness not rejected for GARCH-type forecasts, rejected for Hist-20 (all statistics: [Appendix](#))

 [MFM_ch5_forecast_evaluation](#)

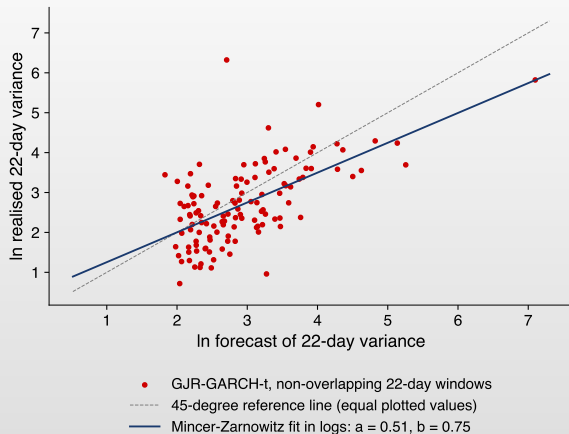
Comparing Forecasting Methods: Giacomini–White and the Model Confidence Set

- With estimated parameters, nested models and an expanding window, the DM differential degenerates under H_0 : DM is not $N(0, 1)$ West (1996), Clark & McCracken (2001)
- Giacomini & White (2006): test the forecasting **method** (model + rolling window of $W = 1\,000$ days), valid for nested models because the estimation noise does not vanish
- Six models, five pairwise tests: a multiple-comparison problem; **Model Confidence Set** Hansen, Lunde & Nason (2011) drops the worst model until equal accuracy is not rejected; the set contains the best model with probability $\geq 1 - \alpha$
- Result: GW rejects only EWMA and Hist-20; the 90% MCS keeps the four GARCH models: the DM verdict, now calibrated (formulas and full table: [Appendix](#))

Model	GW p	MCS p
GARCH-N	0.054	0.586
GARCH-t	–	0.353
GJR-t	0.288	1.000
EGARCH-t	0.201	0.586
EWMA	0.003	0.008
Hist-20	< 0.001	0.001

- GW: rolling window, 2 692 forecasts, vs GARCH-t; MCS: expanding window, QLIKE, stationary bootstrap

One-Month Forecasts: Mincer–Zarnowitz in Logs



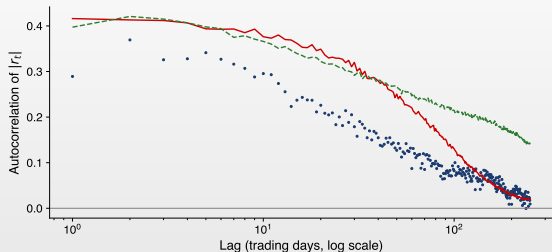
- GJR-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows; R^2 in logs 0.38
- Slope in logs $b = 0.75$ (s.e. 0.07): descriptive only, since $\mathbb{E}[\ln Y | \mathcal{F}] \neq \ln \mathbb{E}[Y | \mathcal{F}]$; $b < 1$ does not prove over-dispersed forecasts
- In levels b is only 0.27: two COVID-19 windows dominate; compare forecasts with QLIKE [Patton \(2011\)](#)

GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987): $r_t = c + \kappa \sigma_t + \varepsilon_t$ (or $\kappa \sigma_t^2$)
 - ▶ $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t ; 0.028 (t = 2.35) with σ_t^2
 - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

 MFM_ch5_long_memory

Long Memory: Where GARCH Falls Short



• S&P 500 data, ACF of $|r_t|$ — Simulated from fitted GARCH(1,1)-t - - - Simulated from fitted FIGARCH(1,d,1)-t

- S&P 500, ACF of $|r_t|$ (log scale): data 0.289, 0.082, 0.009 at lags 1, 100, 250; GARCH(1,1)-t too high at lag 1 (0.416), then an exponential collapse
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996), $(1 - L)^d$: hyperbolic decay; $\hat{d} = 0.60$ (s.e. 0.06), BIC lower by 34.2, but the memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: realised volatility and HAR in Chapter 9

Limits of GARCH and the Extensions They Motivate

Limit	Evidence in this chapter	Extension
Level shifts inflate $\alpha + \beta$ Lamoureux & Lastrapes (1990)	BET-TR (2014–2026) 0.9188 vs BET (2000–2026) 0.9907	GARCH-MIDAS, a slow long-run component Engle, Ghysels & Sohn (2013) ; regime switching Hamilton & Susmel (1994)
Reaction lag of one day	COVID-19: the VIX moved the same day, GARCH the next	Realized GARCH, with a realised-variance measurement equation Hansen, Huang & Shek (2012) (Chapter 9)
One outlier drives the update	EUR/RON residual of 154 (May 2025)	Score-driven (GAS) models with a bounded score Creal, Koopman & Lucas (2013) ; the case study below
Short memory	ACF of $ r_t $ at lags 100–250	FIGARCH; HAR on realised variance (Chapter 9)
One asset at a time	portfolios need covariances	Multivariate GARCH and DCC (Chapter 6)
Managed prices	EUR/RON: long calm, then jumps	Model interventions as jumps or regimes

□ Each extension is judged like GARCH: out of sample, robust loss, Model Confidence Set (model equations: [Appendix](#))

Case Study: Question and Model

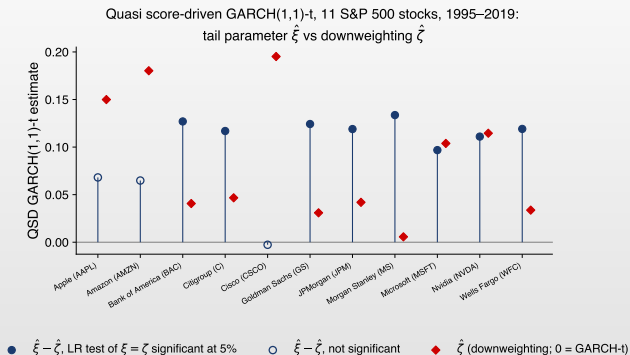
- **Question** Blasques, Francq & Laurent (2023): must a score-driven volatility model update σ_t^2 with the score of the same density that describes the returns?
- Score-driven (GAS) update Creal, Koopman & Lucas (2013): $f_{t+1} = \omega + \alpha S(f_t) \partial \log p(y_t | f_t) / \partial f_t + \beta f_t$; with a Student-t density one ν sets both the tails of z_t and how strongly a large shock is downweighted
- **QSD GARCH(1,1)-t**, eq. (30) of the paper: $y_t = \sqrt{f_t} \epsilon_t$, ϵ_t standardised Student-t with $1/\xi$ degrees of freedom (f_t is σ_t^2 , ϵ_t is z_t)
 - ▶ $f_{t+1} = \omega + \varpi X_t + \alpha \frac{1 + \zeta}{1 - 2\zeta + \zeta \epsilon_t^2} \epsilon_t^2 f_t + \beta f_t$
 - ▶ Two parameters: ξ , the tail thickness of ϵ_t , and ζ , the downweighting of large ϵ_t^2 in the update; $\zeta = 0$: GARCH(1,1)-t; $\zeta = \xi$: Beta-t-GARCH (score-driven)
- Data: 400 S&P 500 stocks, 3 January 1995 (or later) – 28 February 2019; X_t : the squared VIX at a daily horizon; maximum likelihood with an AR(1) mean; LR tests of $\zeta = 0$ and $\xi = \zeta$ against χ_1^2 (full method: [▶ Appendix](#))

Case Study: Key Results (Paper, Tables 4–6)

Rejected at 5%, % of the 400 stocks	Student-t (Table 4)	Skewed t (Table 6)
LR, $H_0: \zeta = 0$ (GARCH update)	89.5	98.8
LR, $H_0: \xi = \zeta$ (score-driven; skewed: also $\kappa = \tau$)	53.5	100.0
PIT test: GARCH(1,1)	40.0	33.1
PIT test: Beta-t-GARCH	46.0	72.2
PIT test: QSD GARCH(1,1)	45.7	11.3

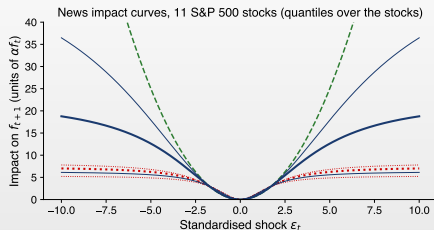
- Downweighting large shocks matters: $\zeta = 0$ is rejected for 89.5% of the stocks and every $\hat{\zeta}$ is positive (Table 4)
- But Beta-t-GARCH downweights too much: where $\hat{\xi} - \hat{\zeta}$ is significant, $\hat{\xi} > \hat{\zeta}$ for all stocks but one; median $1/\hat{\zeta}$ is 9.48, against 5.07 for Beta-t-GARCH ($\zeta = \xi$) (paper, Figs. 2–3): tails and downweighting are two different parameters
- Skewed t: $\hat{\tau}$ far below 0.5 (median 0.333) while $\hat{\kappa}$ stays near 0.5 (Figs. 5–6): leverage enters through the update, not the density; the QSD model then fails the PIT test for only 11.3% of the stocks

Case Study: Replication on 11 Stocks (cf. Paper, Fig. 2)



- Course sample: 11 large stocks still listed (the paper: 400), 1995 (or later) – February 2019, 4986–6081 returns each;
 $X_t = \text{VIX}_t^2 / 252$
- $\zeta = 0$ rejected for 9 of 11 (paper: 89.5%);
 $\xi = \zeta$ rejected for 8 (paper: 53.5%); $\hat{\xi} > \hat{\zeta}$ for 10 of 11
- Median $1/\hat{\zeta}$: 21.4 (QSD) against 6.32 (Beta-t-GARCH); paper: 9.48 against 5.07; 11 selected survivors: imprecise and not representative

Case Study: News Impact Curves (cf. Paper, Fig. 3)



- GARCH(1,1)-t ($\zeta = 0$)
- Beta-t-GARCH, $\zeta = \hat{\zeta}$, median over stocks (0.158)
- Beta-t-GARCH, $\zeta = \hat{\zeta}$, 5% and 95% quantiles (0.139, 0.231)
- QSD GARCH(1,1)-t, $\hat{\zeta}$, median over stocks (0.047)
- QSD GARCH(1,1)-t, $\hat{\zeta}$, 5% and 95% quantiles (0.018, 0.188)

- Weight of ϵ_t in f_{t+1} (units of αf_t) at quantiles of the 11 estimates: updating functions at a common αf_t , not fitted paths
- A 5-s.d. shock adds 25 (GARCH-t), 12.6 (median QSD), 6.2 (median Beta-t-GARCH); at 10 s.d.: 100, 18.8, 7.0 (bound $(1 + \zeta)/\zeta = 7.3$)
- QSD lies between the two: large shocks are downweighted, far less than Beta-t-GARCH imposes

An Open Question and the Discovery Loop

- Do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample, once the number of models compared is accounted for?
 - ▶ Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models; Hansen, Lunde & Nason (2011): the Model Confidence Set is the right object when many models compete
 - ▶ Formal: $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$, $H_0: \mathbb{E}d_t = 0$; falsified if A leaves the Model Confidence Set, or if the sign of $\bar{d}$ flips across pre-registered subperiods
- Loop: an AI assistant drafts the literature map, competing hypotheses, code, a robustness plan and a referee-style critique; the researcher resolves every DOI, reproduces every number and fixes the plan before the first run
- Checks specific to volatility forecasts
 - ▶ Look-ahead (h_t must not move when r_t changes); proxy-robust loss (QLIKE, MSE); model mining (Model Confidence Set or Bonferroni); scale and convergence of the optimiser; invented DOIs or arch arguments
 - ▶ Reusable prompts and the full checklist: [▶ Appendix](#)

Mini-Case and a Team-Project Seed

- A typical assistant claim: “GJR-t beats GARCH-t because of the leverage effect”; $d_t = \text{QLIKE}(\text{GJR-t}) - \text{QLIKE}(\text{GARCH-t})$, 2 692 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$
- Testable implication of leverage: the gain sits after down days
 - ▶ After $r_{t-1} < 0$ (1 223 days): $\bar{d} = -0.015$, $t = -0.51$; after $r_{t-1} \geq 0$ (1 469): $\bar{d} = -0.034$, $t = -1.12$: not supported; 2016–2020 $t = -0.02$, 2021–2026 $t = -4.06$
 - ▶ Descriptive only (nested models, expanding window): the calibrated follow-up is a fixed-window GW test of the sign difference, with d_t regressed on $1(r_{t-1} < 0)$
- **Project seed:** do machine-learning forecasts enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
 - ▶ Replicate the QLIKE table first; pre-register models, losses, splits and error rate; log AI use in AI_USE.md and every caught error in AI_ERRORS.md (full brief: [▶ Appendix](#))

Synthesis: a Decision Workflow

- ▣ 1. **Detect:** $Q(10)$ on r_t^2 and the LM test; clustering is universal, persistence $\alpha + \beta$ is close to 1 in all our markets
- ▣ 2. **Fit:** GARCH, GJR, EGARCH with Normal, Student-t, skewed-t innovations; returns in %; check convergence and parameters at a bound
- ▣ 3. **Diagnose:** level, squares and signs of $\hat{z}_t$ and its largest values; naive tests are conservative
- ▣ 4. **Forecast:** the recursion gives multi-step variances; the square-root-of-time rule errs in the direction of mean reversion
- ▣ 5. **Compare:** out of sample, QLIKE, GW with a rolling window and the Model Confidence Set: GARCH beats EWMA and Hist-20, the GARCH family is hard to separate
- ▣ 6. **Report uncertainty:** robust s.e. (valid if $\mathbb{E}z_t^4 < \infty$), boundary-aware tests, half-lives as intervals $[h_{lo}; \infty)$ when $\alpha + \beta \approx 1$

Check Yourself, and Next: Chapter 6

Retrieval questions

- Why does the square-root-of-time rule overstate the 10-day risk after a turbulent day?
- Why is the LR test of Normal against Student-t innovations compared with 2.71 and not 3.84?
- Why can a symmetric interval for the half-life contain negative values?
- Why is a gap between the VIX and GARCH volatility not a measured variance risk premium?

Next chapters

- **Chapter 6:** multivariate volatility and dynamic correlations (DCC)
- **Chapters 7–8:** Value at Risk and expected shortfall with GARCH volatility, and their backtesting
- **Chapter 9:** realised volatility and long memory
- **Seminar 5:** the variance recursion step by step, GARCH-t and GJR on real data, an AI answer to audit, bootstrap tests and forecast comparisons

Appendix: Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4
BET	Bucharest Stock Exchange	2000	6 670	249.8
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3
EUR/RON reference rate	BNR	2005	5 352	252.4
Gold (XAU/USD)	spot market	2000	6 922	259.2

- Daily closes up to 18 September 2026; returns $r_t = 100 (\ln P_t - \ln P_{t-1})$, in %
- Equity indices and gold: weekdays only; gold volatility is annualised using its actual observation frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

 MFM_ch5_arch_effects

Appendix: ARCH Effects, All Statistics

Series	Vol. (% p.a.)	Excess kurt.	$\hat{\rho}_1(r^2)$	$Q(10)$ on r^2	LM(5)
S&P 500	19.2	10.7	0.313	6133	1684
BET	22.1	11.1	0.330	2440	982
BET-TR	15.4	17.8	0.171	714	371
Bitcoin	66.6	12.0	0.137	213	115
EUR/RON	4.4	15.1	0.342	2420	871
Gold	17.6	7.7	0.110	744	329

□ 5% critical values: χ^2_{10} : 18.3; χ^2_5 : 11.1; all p-values < 0.001

 MFM_ch5_arch_effects

Appendix: One-Sided Score Tests and the Davies Problem

- The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
 - ▶ [Lee & King \(1993\)](#): one-sided score test built on the sum of the score components, $N(0, 1)$ under H_0 , rejects in the upper tail only
- Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
 - ▶ The ARCH(q) LM statistic is also the score test against GARCH(p, q) [Lee & King \(1993\)](#): the same statistic serves both alternatives

Appendix: Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

BY-SA 4.0; Wikimedia Commons



Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC

BY-SA 2.0; Wikimedia Commons

- Nobel Prize 2003: Engle "for methods of analyzing economic time series with time-varying volatility (ARCH)"
 - Shared with Clive Granger (cointegration)
- Nobel lecture: [Engle \(2004\)](#), on risk and volatility in financial practice
- Tim Bollerslev, Engle's doctoral student, generalised ARCH to GARCH in 1986

Appendix: Kurtosis: Rounding and a Long Simulated Path

- S&P 500, GARCH-N: the moment condition is 0.990435, so $K = 12.1$ has the denominator $1 - 0.990435$
 - ▶ Rounding to $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$ gives 0.9916 and $K = 13.5$: report such functions from unrounded estimates
- Simulated GARCH(1,1) of the simulation slide, one million days
 - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2 : 0.26
 - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters

Appendix: QMLE: Variants of the Theorem

- ▣ Estimated mean: the score gains the term ε_t/σ_t^2 and the covariance $(\kappa_z - 1)J^{-1}$ changes form
- ▣ IGARCH ($\alpha + \beta = 1$): strict stationarity still holds, and the QMLE remains consistent and asymptotically Normal [Lumsdaine \(1996\)](#)
- ▣ Explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) \geq 0$: [Jensen & Rahbek \(2004\)](#) cover (α, β) with ω fixed, since ω is not identified
- ▣ Heavy tails, $\kappa_z = \infty$: slower than $\sqrt{n}$, non-Gaussian limits [Hall & Yao \(2003\)](#); heavy-tailed quasi-likelihoods with a scale adjustment [Fan, Qi & Xiu \(2014\)](#)

Appendix: The Li-Mak Correction for Squared-Residual Autocorrelations

- Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n} \hat{r} \rightarrow N(0, C)$, $C = I - H' J^{-1} H / (\kappa_z - 1)$ Li & Mak (1994)
 - ▶ $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$: how the fitted variance absorbs lag- k dependence; the ARMA rule χ_{m-p-q}^2 does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
 - ▶ The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence; corrected statistic $n \hat{r}' C^{-1} \hat{r} = 19.1$

 MFM_ch5_inference

Appendix: S&P 500 Forecast Evaluation, All Statistics

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	-1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	—	0.88	0.238	0.80
GJR-t	0.732	21.8	-1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	-0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ: $r_t^2 = a + b h_t$; Wald p: test of $a = 0$, $b = 1$

 MFM_ch5_forecast_evaluation

Appendix: Giacomini–White and the MCS: Formulas and Full Table

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.751	1.87	5.85	0.054	0.586
GARCH-t	0.743	—	—	—	0.353
GJR-t	0.738	-0.18	2.49	0.288	1.000
EGARCH-t	0.746	0.07	3.21	0.201	0.586
EWMA	0.815	3.20	11.40	0.003	0.008
Hist-20	0.889	3.87	16.41	< 0.001	0.001

- Conditional test $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = 0$ with instruments $x_t = (1, d_t)$: $GW = n \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2_2$, $Z_{t+1} = x_t d_{t+1}$ (critical value 5.99)
- Rolling window $W = 1000$ days, re-estimated each January, 2692 forecasts; tests against GARCH-t, HAC with 5 lags
- MCS on the expanding-window forecasts: QLIKE, statistic T_R , stationary bootstrap (mean block 10, 10 000 draws); 90% MCS: GARCH-N, GARCH-t, GJR-t, EGARCH-t

Appendix: Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma_{t+1}^2 = \omega + A s_t + B \sigma_t^2$, s_t the scaled score of the Student-t density
 - ▶ The score is bounded in z_t : an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ_t^2 to a realised variance
 - ▶ Uses intraday information and reacts faster; the bridge to [Chapter 9](#)
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma_t^2 = \tau_t g_t$, a slow long-run component τ_t driven by macro data (mixed frequencies)
 - ▶ A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1

Appendix: Beyond Variance: Entropy and Tail Risk

- [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
 - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): the entropy of intraday Bitcoin returns proposed as an additional signal for daily Value at Risk
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- GARCH volatility feeds directly into Value at Risk and expected shortfall: [Chapters 7–8](#)

Appendix: Case Study, Full Method

- **Why it matters:** GARCH is driven by ε_t^2 whatever the density, so an isolated extreme shock can produce an excessive volatility response lasting days or weeks (Section 1 of the paper)
 - ▶ With a left-skewed density, a pure score-driven model raises volatility more after positive returns: the opposite of the leverage effect (Example 2)
 - ▶ QSD models, eq. (3): $f_{t+1} = \omega + \alpha \psi(y_t, X_t, f_t, \theta) + \beta f_t$, ψ the scaled score of an updating function ρ that need not equal $\log \rho$; theory: consistency, asymptotic normality, LR and LM tests with χ^2 limits (Section 3)
- **Estimation:** $\zeta = 0$ is kept inside the parameter space by eqs. (31)–(32): $\zeta \in (-1, 1/2)$ and $\Psi(x) = x(1 - e^{-cx})/(1 + e^{-cx})$ applied to the weight, $c = 1000$
 - ▶ PIT test [Diebold, Gunther & Tay \(1998\)](#): $u_t = F(\epsilon_t)$ i.i.d. $U(0, 1)$ if the model is right; with the MLE and [Pierce \(1982\)](#), reject if $|\sum_t \hat{u}_t - T/2|/\sqrt{T(1/12 - B\mathcal{I}(\hat{\theta})^{-1}B^\top)} > 1.96$, $B = \lim_{T \rightarrow \infty} \mathbb{E}[\partial \sum_t (u_t - T/2)/\partial \theta^\top]$, $\mathcal{I}$ the Fisher information; it detects a shifted PIT mean, not every departure from uniformity
- **Data:** stocks with fewer than 4000 observations or with convergence problems are dropped: 400 stocks (Section 5)
 - ▶ Skewed variant, eq. (47): a skewed Student-t with asymmetry κ for ϵ_t and separate (ζ, τ) in the update; $\tau < 0.5$ gives a leverage effect whatever κ (Section 5.2)

Appendix: Case Study, an Example and the Limits Stated by the Authors

- Centene, 4330 days: $\hat{\xi} \approx 0.26$ (about 4 degrees of freedom) in all three models, but $\hat{\zeta} = 0.048$ (s.e. 0.027); LR 8.81 ($p = 0.003$) for $\xi = \zeta$ and 8.42 ($p = 0.004$) for $\zeta = 0$ (Table 5); after a crash day in March–April 2008, GARCH-t volatility stays well above Beta-t-GARCH and QSD lies between (Fig. 4)
- **Limits** stated by the authors
 - ▶ Beating GARCH-t and Beta-t-GARCH does not make the Student-t QSD model adequate: the three models fail the PIT test for 40.0–46.0% of the stocks (Table 4)
 - ▶ In the unmodified recursion (30), $\zeta < 0$ or $\zeta > 1/2$ does not define a proper volatility model, and $\zeta = 1/2$ with $\varpi = 0$ is degenerate (Section 4.1)
 - ▶ The $N(T/2, T/12)$ law of $\sum_t u_t$ holds only for known parameters; with estimated ones the PIT test needs the correction above (Section 5.1)
- **Open extensions**
 - ▶ Does the bounded QSD update beat GJR-GARCH-t in QLIKE out of sample, especially after March 2020? Does it tame the EUR/RON residual of 154? (Seminar 5, C3)

Appendix: The Discovery Loop: Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
 - ▶ Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy [Patton \(2011\)](#)
- **Hypotheses:** *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code:** *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from the course daily price files and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
 - ▶ Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM $t = -1.22$

Appendix: The Discovery Loop: Reusable Prompts (2/2)

- **Robustness:** *“Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction.”*
 - ▶ A random-split placebo is calibrated only if it reassigns blocks of days, preserving volatility clustering
- **Critique:** *“Act as a referee: could the proxy r_t^2 , one crisis year or the HAC lag create my result? Give the test that would expose each.”*
- **Report:** all splits and losses tried, adjusted p -values, code, seeds and package versions; a specification chosen after seeing results is exploratory and reported as such
- **Influential periods:** a crisis year can dominate $\bar{d}$; report subperiods and state in advance whether any year may be excluded

Appendix: The Team-Project Seed in Full

- Question: do machine-learning forecasts on daily data enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
- Realised-variance (HAR) branch: the course data set of 5-minute realised variances holds 40 large American stocks, January 2015 – January 2026; there are no intraday data for the S&P 500 index or the BET, so this branch is run stock by stock
- Replicate first: the QLIKE table (GARCH- t 0.757, EWMA DM $t = 3.48$) and the mini-case splits
- Deliverables
 - ▶ A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
 - ▶ Code on the course market data, the Model Confidence Set at 10%, a referee-style self-critique; `AI_USE.md` and `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_05/ai_discovery_case.py](#)

Appendix: Further Resources

▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

▣ Code

- ▶ All charts and tables: [Quantlets/Ch_05](#); Python package [Sheppard, arch](#)

References (1/6)

- ▣ Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- ▣ Andrews, D.W.K. (2000). Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2), 399–405.
- ▣ Andrews, D.W.K. (2001). Testing when a parameter is on the boundary of the maintained hypothesis. *Econometrica*, 69(3), 683–734.
- ▣ Baillie, R.T., Bollerslev, T., Mikkelsen, H.O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30.
- ▣ Berkes, I., Horváth, L., Kokoszka, P. (2003). GARCH processes: structure and estimation. *Bernoulli*, 9(2), 201–227.
- ▣ Blasques, F., Francq, C., Laurent, S. (2023). Quasi score-driven models. *Journal of Econometrics*, 234(1), 251–275.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.

References (2/6)

- ▣ Christensen, K., Siggaard, M., Veliyev, B. (2023). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5), 1680–1727.
- ▣ Clark, T.E., McCracken, M.W. (2001). Tests of equal forecast accuracy and encompassing for nested models. *Journal of Econometrics*, 105(1), 85–110.
- ▣ Creal, D., Koopman, S.J., Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.
- ▣ Diebold, F.X., Gunther, T.A., Tay, A.S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Engle, R.F. (2004). Risk and volatility: econometric models and financial practice. *American Economic Review*, 94(3), 405–420.
- ▣ Engle, R.F., Ghysels, E., Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *Review of Economics and Statistics*, 95(3), 776–797.
- ▣ Engle, R.F., Lilien, D.M., Robins, R.P. (1987). Estimating time varying risk premia in the term structure: the ARCH-M model. *Econometrica*, 55(2), 391–407.

References (3/6)

- ▣ Engle, R.F., Ng, V.K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778.
- ▣ Fan, J., Qi, L., Xiu, D. (2014). Quasi-maximum likelihood estimation of GARCH models with heavy-tailed likelihoods. *Journal of Business & Economic Statistics*, 32(2), 178–191.
- ▣ Francq, C., Zakoïan, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637.
- ▣ Francq, C., Zakoïan, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications*, 2nd ed. Wiley.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Glosten, L.R., Jagannathan, R., Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.
- ▣ Hall, P., Yao, Q. (2003). Inference in ARCH and GARCH models with heavy-tailed errors. *Econometrica*, 71(1), 285–317.
- ▣ Hamilton, J.D., Susmel, R. (1994). Autoregressive conditional heteroskedasticity and changes in regime. *Journal of Econometrics*, 64(1–2), 307–333.

References (4/6)

- ▣ Hansen, B.E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730.
- ▣ Hansen, P.R., Huang, Z., Shek, H.H. (2012). Realized GARCH: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906.
- ▣ Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ J.P. Morgan/Reuters (1996). RiskMetrics — Technical Document, 4th ed. New York.
- ▣ Jensen, S.T., Rahbek, A. (2004). Asymptotic inference for nonstationary GARCH. *Econometric Theory*, 20(6), 1203–1226.
- ▣ Lamoureux, C.G., Lastrapes, W.D. (1990). Persistence in variance, structural change, and the GARCH model. *Journal of Business & Economic Statistics*, 8(2), 225–234.
- ▣ Lee, J.H.H., King, M.L. (1993). A locally most mean powerful based score test for ARCH and GARCH regression disturbances. *Journal of Business & Economic Statistics*, 11(1), 17–27.
- ▣ Li, W.K., Mak, T.K. (1994). On the squared residual autocorrelations in non-linear time series with conditional heteroskedasticity. *Journal of Time Series Analysis*, 15(6), 627–636.

References (5/6)

- ▣ Lumsdaine, R.L. (1996). Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. *Econometrica*, 64(3), 575–596.
- ▣ Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- ▣ McLeod, A.I., Li, W.K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4(4), 269–273.
- ▣ Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (ed.), *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER.
- ▣ Nelson, D.B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.
- ▣ Nelson, D.B. (1991). Conditional heteroskedasticity in asset returns: a new approach. *Econometrica*, 59(2), 347–370.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). Modeling expected shortfall using tail entropy. *Entropy*, 21(12), 1204.

References (6/6)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily Value at Risk. *Entropy*, 21(2), 102.
- ▣ Pierce, D.A. (1982). The asymptotic effect of substituting estimators for parameters in certain types of statistics. *The Annals of Statistics*, 10(2), 475–478.
- ▣ Self, S.G., Liang, K.-Y. (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398), 605–610.
- ▣ Sheppard, K. [arch: ARCH models and other tools for financial econometrics in Python](#) (software documentation).
- ▣ West, K.D. (1996). Asymptotic inference about predictive ability. *Econometrica*, 64(5), 1067–1084.